

Pregame talks

Math AA HL

Sprint 3 · Thursday: Topic 1 remembered, plus 2.2 and 2.12
The numbers nobody wanted

Time limit: 20 minutes, strict. Start a timer. Stop when it rings, finished or not. Answers are on page 2: use them if you are stuck, and say so in the email. The point is to try.

What to send, by Friday 9 October, 20:00: one email to Gauti.Eggertsson@gmail.com, subject "Sprint 3", three lines:

Done at [time]. Got to problem [n]. Looked at answers for [which, or none].

On the weekend you redo these with Gauti over Zoom, without the answers.

1. Solve $x^2 - 2x + 5 = 0$. Write the two roots, check that they are a conjugate pair, and check Vieta: their sum and their product.

2. $2 + i$ is a root of $x^2 + bx + c = 0$, where b and c are real. Find b and c without solving anything: use the conjugate pair and Vieta.

3. The sequence where each term is the previous term minus the one before it: $u_{n+1} = u_n - u_{n-1}$, with $u_0 = 1$, $u_1 = 1$. (a) Write the first nine terms. (b) After how many steps does it repeat? (c) Solve its characteristic equation $r^2 - r + 1 = 0$ and find $|r|$.

4. Topic 1, remembered. The geometric sequence 3, 6, 12, 24, ... (a) Find the 10th term. (b) Find the sum of the first 10 terms.

Look only when stuck, and say which ones in the email. On the weekend these are redone without this page.

1. $\Delta = 4 - 20 = -16$. $x = \frac{2 \pm 4i}{2} = 1 \pm 2i$. Conjugates. Sum $= 2 = -\frac{-2}{1}$, product $= (1 + 2i)(1 - 2i) = 1 + 4 = 5 = \frac{5}{1}$.
2. The other root is $2 - i$. Sum $= 4 = -b$, so $b = -4$. Product $= (2 + i)(2 - i) = 5 = c$. The quadratic is $x^2 - 4x + 5$.
3. (a) 1, 1, 0, -1, -1, 0, 1, 1, 0. (b) Every 6 steps. (c) $r = \frac{1 \pm i\sqrt{3}}{2}$, and $|r| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$. Two imaginary roots on the unit circle, and the sequence is a wave with period 6.
4. $u_1 = 3$, $r = 2$. (a) $u_{10} = 3 \cdot 2^9 = 1536$. (b) $S_{10} = 3 \cdot \frac{2^{10} - 1}{2 - 1} = 3 \cdot 1023 = 3069$.